

Waltenhofen's Pendulum

P&S: Magnetische Felder und Wellen im Alltag

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Abstract

We built *Waltenhofen's pendulum*, in order to show the effect of eddy currents. For this we used non-ferromagnetic and conducting materials like *aluminium* and *copper*. By swinging different shapes and sizes of material through a *strong magnetic field* one can show the force induced by eddy currents. This enabled us to measure the time until the pendulum stops swinging and observe expected results.



Figure 1: Setup for experiment.

1. Introduction

Eddy current brakes are commonly used for different applications. Vehicles, circular saws, and many more. Therefore it is interesting to observe its effects in a controlled environment and compare the experimental results to the theory.

2. Theory

The magnetic field of the permanent magnet induces eddy currents into the paramagnetic material, which in turn create a magnetic field and therefore also an opposing force, slowing down the material.

2.1. Lenz's law

Faraday's Law of Induction is given by:

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

where:

- $\mathcal{E}$ is the induced electromotive force (EMF),
- Φ_B is the magnetic flux, given by $\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}$,
- $\mathbf{B}$ is the magnetic field,
- $d\mathbf{A}$ is the infinitesimal area vector perpendicular to the loop's surface.

We see a change in the magnetic field will also create an electromagnetic force. The minus sign at the beginning shows that this force works the opposite as its cause. Otherwise we would get a positive feedback loop with infinite energy that is physically impossible. This phenomenon is known as Lenz's law.

2.2. Eddy current

Eddy currents are loops of electric current, induced by a changing magnetic field, or the relative motion of a conductor in a constant magnetic field. They flow in closed loops, in a plane perpendicular to the magnetic field.

If this change in the magnetic field is caused by the movement of these electric conductors, the energy to create these currents is "taken away" from the kinetic energy of such conductors. In other words they are slowed down. This is in accordance with Lenz's law, as the currents work against their cause, that being the mechanical motion of the conductors.

But what happens with this energy? It is converted into heat due to the inner resistance of the different materials.

The energy loss due to heat in the internal resistance of a source is given by:

$$P = I^2 r$$

where:

- P is the power dissipated as heat (in watts),
- I is the current flowing through the circuit (in amperes),
- r is the internal resistance of the source (in ohms).

Over time, the total energy lost as heat is:

$$E = I^2 r t$$

where t is the duration of time the current flows (in seconds).

2.3. Materials

During our experiments we used plates made out of copper and aluminium. Here is a brief summary of their resistivities, that can be used to calculate the energy loss.

Material	Resistivity (ρ) ($\Omega \cdot \text{m}$)	Conductivity
Copper	1.68×10^{-8}	100%
Aluminum	2.82×10^{-8}	61%

Table 1: Comparison of resistivity and conductivity of copper and aluminum.

2.4. Eddy current brakes

A way to benefit from this physical phenomenon is by building eddy current brakes. Having two conducting plates slide next to each other, they will cause such eddy currents and slow each other down. The bigger the area, the higher the rate of deceleration.

Why would you use such a system in real life? As no contact is required to create a braking effect, you can avoid any mechanical wear, thereby creating very low maintenance brakes. They are also reliable, as you can calculate the amount of braking by looking at the properties of the materials used, which is much harder by brakes taking advantage from friction, as they tear and change their properties over time.

Problems can also arise from this system, such as its speed dependency. It may take long time to actually bring a system to halt.

Non-Destructive Testing (NDT), Induction Heating, Magnetic Levitation (Maglev), Industrial saws or Roller Coasters are some real world applications that take advantage from this technology.

2.5. Waltenhofen's Pendulum

The Waltenhofen's Pendulum is a pendulum where you put magnets on the pendulum, and also on the sides of it parallel to the direction of the oscillation. The amplitude of this pendulum is decreased by eddy currents on magnetic plates used in the experiment. Our project was the construction of such a pendulum, as well as conducting measurements on it.

3. Method and Equipment

We replicated the experimental setup of *Waltenhofen's Pendulum*. For this we created different shapes from aluminium and copper sheets of varying thicknesses. In the original experiment the magnetic field is created by a stationary electromagnet. We decided to use a permanent magnet, attached to the pendulum and stationary sheets. This allowed us to fix the plates without using steel screws and nuts in the magnetic field, slowing down the pendulum and interfering with our results.

3.1. Pendulum and Magnet

For the pendulum we laser-cut 2 plates out of 3mm MDF and attached them to a stand provided to us. The magnets are attached to the arms by magnetic force. Their magnetic field is strong enough to secure them, when placed on opposite sides of the arms. In order to hold the different plates we built 2 stands, where the magnets can be fixed by screws.

We decided on this method due to some unforeseen circumstances. In the beginning we wanted to use an electromagnet, in order to be able to change the magnetic field dynamicaly and turn it off completely. We made it from a transformer, where we cut 2 slits and removed one spool in order to have a strong magnetic field inside the slits. This, however, did not work properly, since the limited current by our power source was too weak to create a proper field.

This led to us using permanent magnets and placing them on the pendulum instead of the plates, since we didn't find a feasible way to attach the plates to our pendulum, without having any ferromagnetic materials interfering with our experiment.

3.2. Plates

In order to properly observe the desired effects we used plates of different thicknesses and materials and crafted different shapes, some of which hinder the induction of eddy currents, thus lowering the effect.

One of the shapes is just a full plate, to use as a control for the other shapes. Two plates have vertical slits, splitting the area of the plate, while the remaining two have a circular hole, one of which is slitted, as seen in fig. 2.



Figure 2: Aluminium plates with different shapes.

For the comparison between materials and thicknesses we used bigger plates, to enhance the force. We used aluminium and copper of thicknesses as seen in table 2. We also tested the significance of the distance between the plate and the magnet with the 2mm aluminium plate and distances specified in table 2.

Aluminium (mm)	Copper (mm)	Distances (mm)
0.1	1.5	10
2	-	20
6	-	30

Table 2: Used thicknesses.

3.3. Experiments

We decided to release the pendulum at an angle of 90° and measure the time until the pendulum's swing was lower than a certain threshold. We repeated this 5-10 times for each individual plate and averaged the results to mitigate differences due to other factors, like friction.

At first we conducted the experiment without any plates, to find the stop-time of the pendulum without any brake forces. Then we measured the times for the 2mm aluminium plate at different distances as seen in table 2. Afterwards we used the aluminium plates of different thicknesses from table 2, to see how that affects the results. We also tested the 1.5mm copper plate, however the results have to be compared carefully, since the thickness, shape and outer dimensions of the plate are different from all the aluminium plates. Lastly we compared the five smaller plates from fig. 2.

4. Results

All in all the result matched our expectations, however there are some discrepancies in the data. We think those mostly arose due to friction and inaccuracies in our timing.

4.1. Change of Distance

First we want to discuss the results of the experiment with varying distances. In fig. 3 you can see the effect of the distance. We stopped after 30mm, since no significant difference to the data of the no plate experiment can be observed for larger distances.

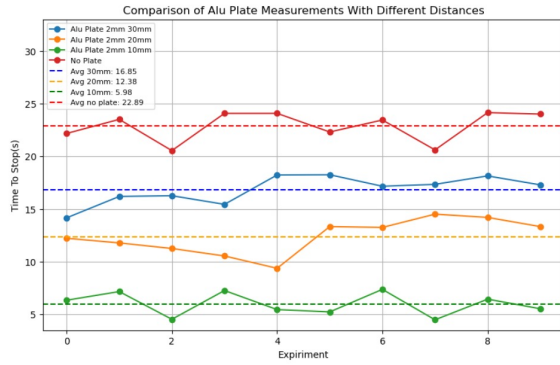


Figure 3: Comparison of Alu Plate Measurements with Different Distances.

4.2. Effect of Plate Thickness

As seen in fig. 4 the braking force does not significantly increase for the 6mm plate, compared to the 6mm plate. The aluminium plate, however does not experience a large slow down. This suggests that there might be a minimal thickness, required to achieve any reasonable force.

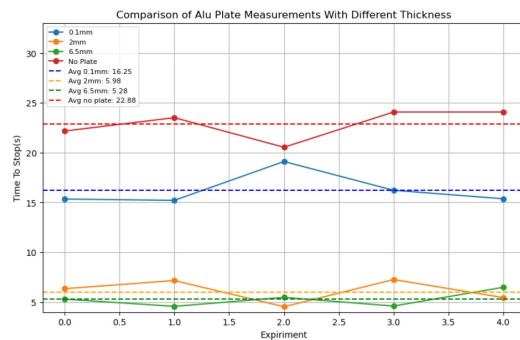


Figure 4: Comparison of Alu Plate Measurements With Different Thicknesses

4.3. Differences in Materials

As mentioned above, the choice of material is important. The main property, which increases the effect, is the conductivity of the non-ferromagnetic material. To observe this, we compared aluminium and copper. The results from the experiment, as seen in fig. 5 match our expectation, since the conductivity of copper is $59.6 \times 10^6 S/m$, compared to $37.7 \times 10^6 S/m$ of aluminium. This enables stronger Eddy Currents, which, in turn, increase the braking force. However, the results of this experiment should be taken with a grain of salt, since we didn't have access to a copper plate of comparable shape and thickness.

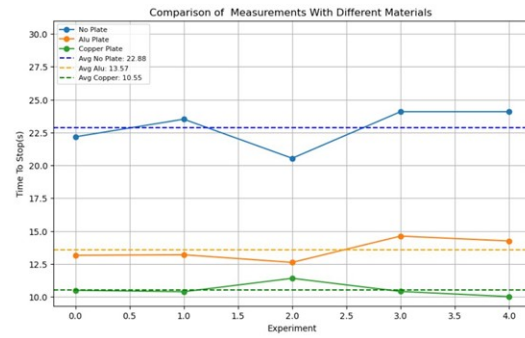


Figure 5: Comparison of Measurements With Different Materials

4.4. Slitted Plates

For this experiment we compared aluminium plates of equal size, one without slits, one with 1 slit and one with 3 slits, as seen in fig. 2. We expect the limited area, due to the slits, hinders the creation of Eddy Currents and therefore lowers the effect. This assumption is true and can be seen in fig. 6.

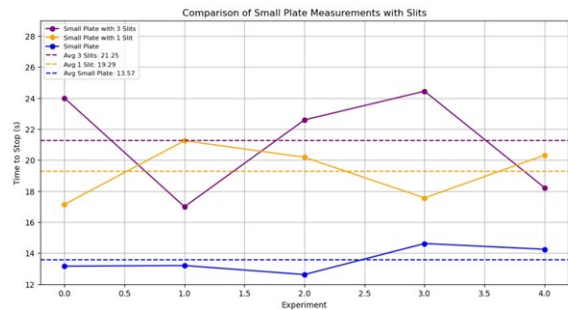


Figure 6: Comparison of Small Plate Measurements with Slits

4.5. Plates with Holes

Lastly we used plates of the same dimensions as in the previous experiment, this time, however, we drilled holes in two of those plates, and sawed a slit in one of those, as seen in fig. 2. Here we expect a lessened effect, in comparison to the full plate, since there is less area to form Eddy currents. Furthermore the slit in the hole will stop Eddy currents to flow around the hole, which will decrease the braking force even further, compared to the plate with the unslitted hole. Those received results match our expectations and can be seen in fig. 7.

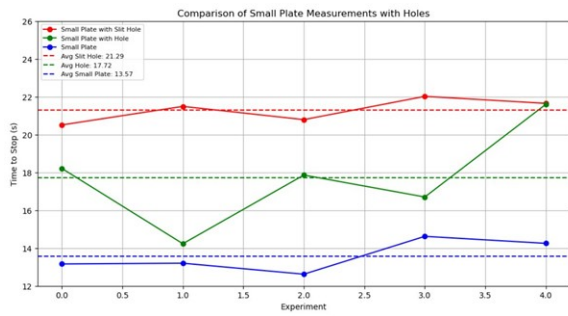


Figure 7: Comparison of Small Plate Measurements With Holes

5. Summary

We built a pendulum to observe the Eddy Currents, slowing down the pendulums movements. The received results mostly match our intuitions about magnetic fields. Eddy Current Brakes look like magic, the pendulum slows down for seemingly no reason, however this exciting phenomenon can be explained by physics. Further interesting work could be done by properly testing different materials, of equal dimensions, and also by use of an adjustable electromagnet.

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